STUDENT PERFORMANCE: EDUCATIONAL EQUITY & INSIGHTS

REPORT

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INTRODUCTION

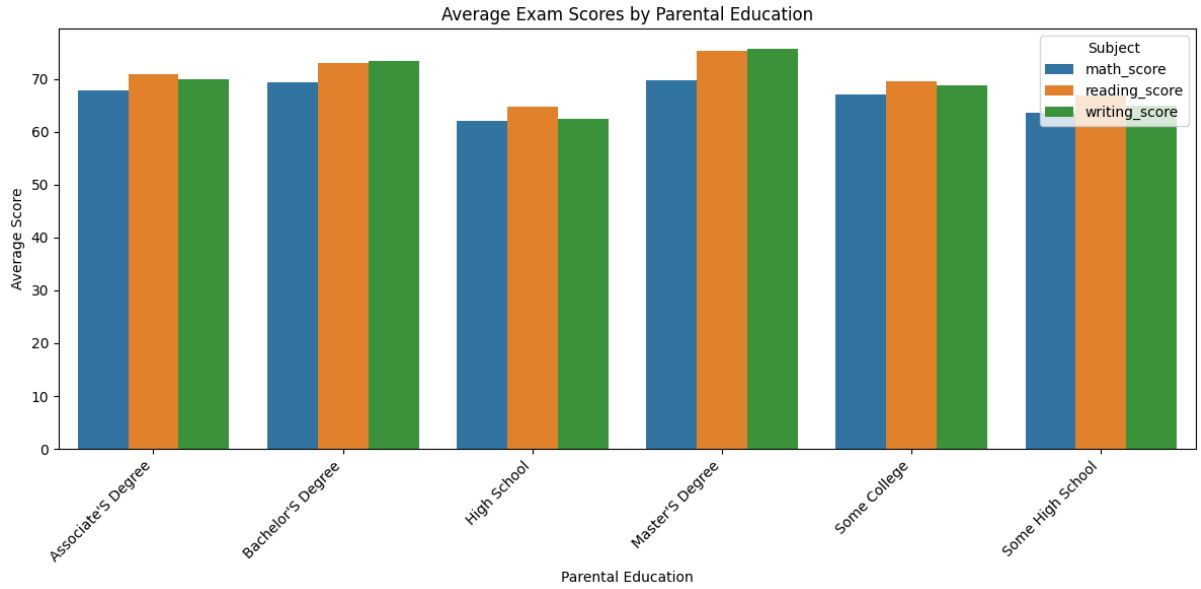
Student performance analysis is an important process in the education system. This report focuses on analyzing students' academic performance using the Students Performance Dataset.

The dataset contains information about students, including gender, parental level of education, lunch type, test preparation course, and marks obtained in mathematics, reading, and writing.

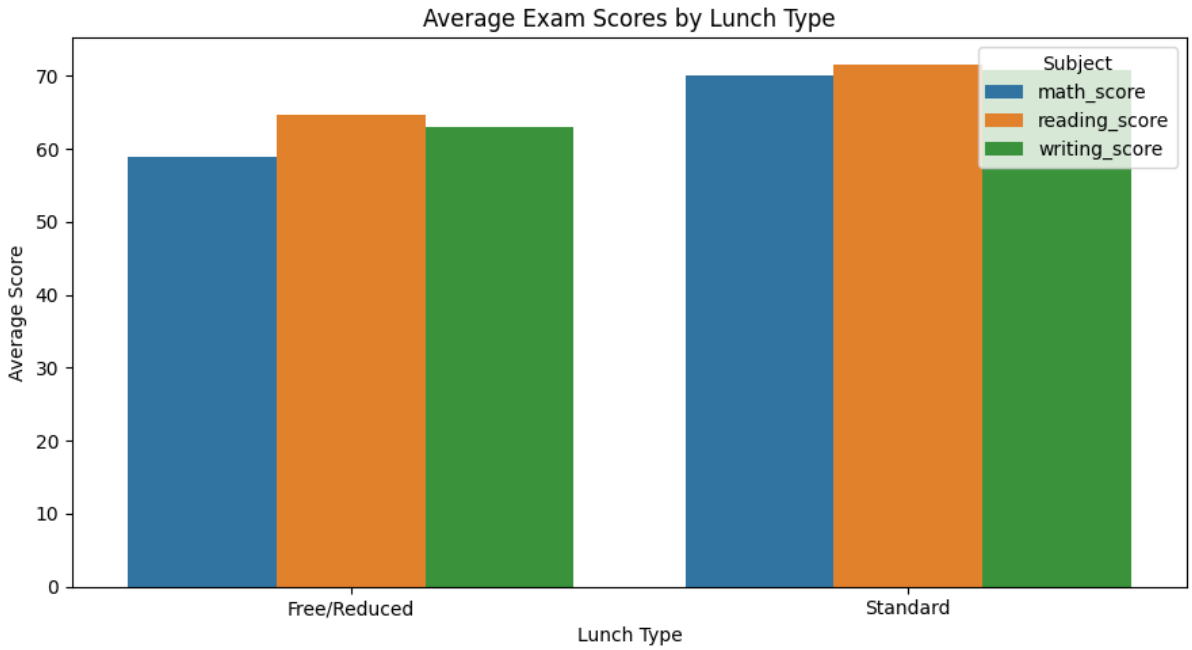
This report helps to understand students' academic performance, compare subject-wise scores, and identify the factors that may influence their results. Data analysis and visualization techniques are used to identify patterns and meaningful insights.

The main aim of this report is to understand student performance and provide useful insights for improving the quality of education.

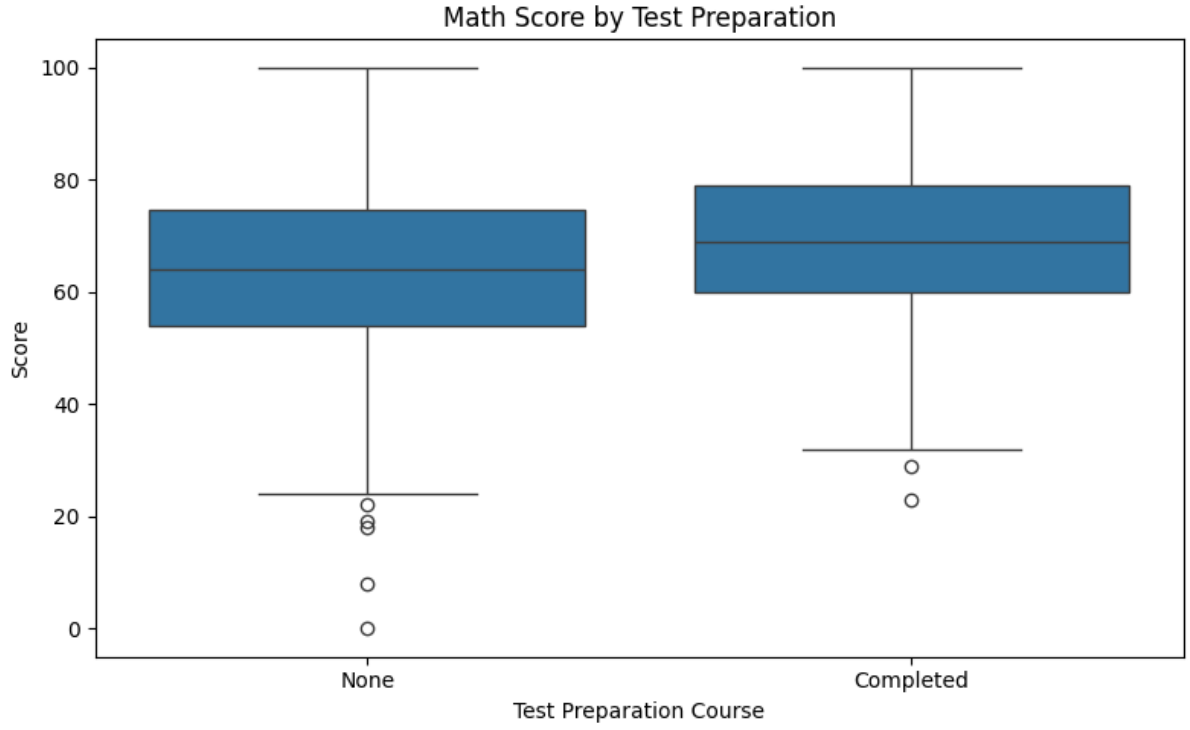
1.Analysis of Parental Education 2 Parental education level is widely documented as an environmental determinant in student outcome variances. The data reveals a consistent positive trend: students whose parents hold post-secondary degrees (Master's or Bachelor's) systematically achieve higher mean scores across all three subjects compared to those whose parents completed high school or some high school.

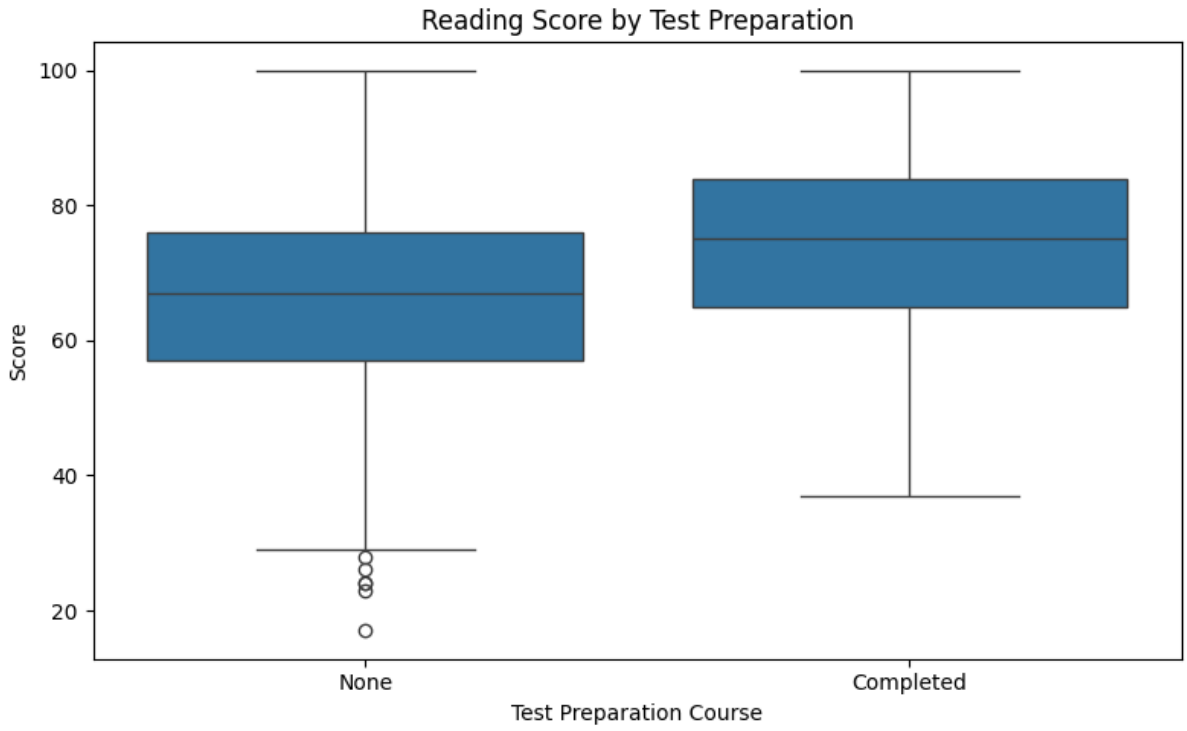


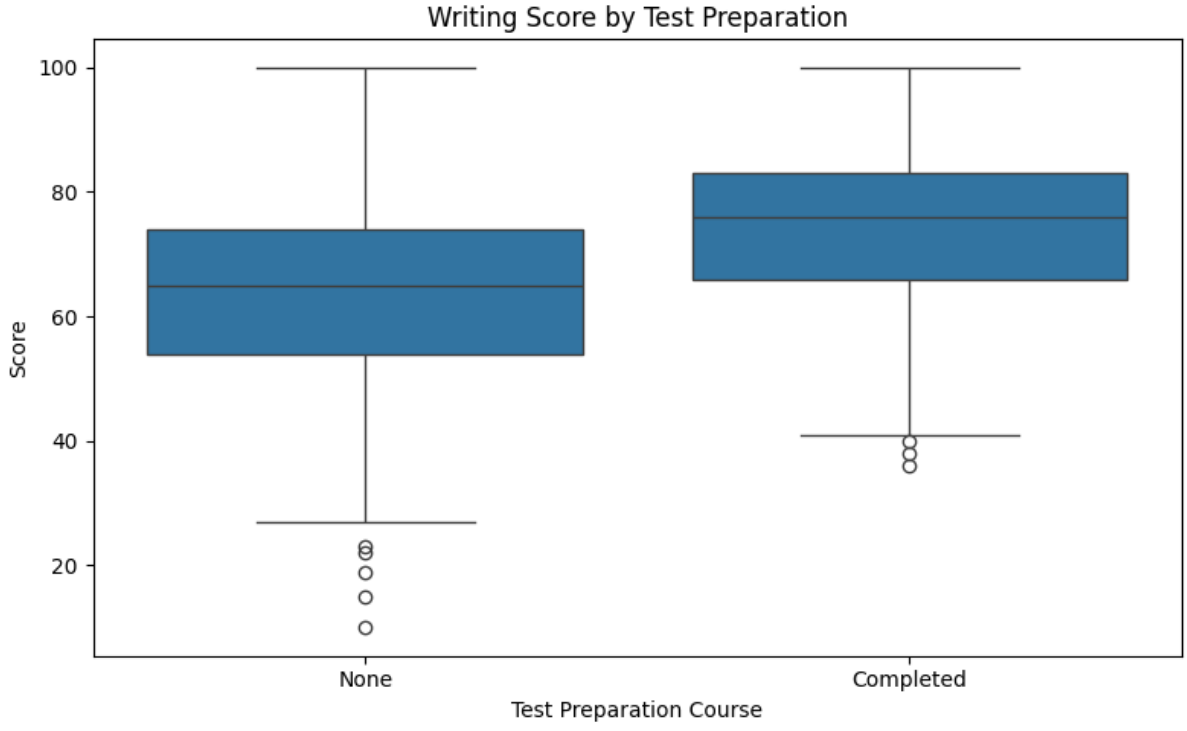
2.Analysis of Lunch Type Lunch subscription type (Standard vs. Free/Reduced) serves as a proxy indicator for household socioeconomic background. A substantial baseline discrepancy exists between the two categories across all academic domains. Students receiving standard lunch demonstrate higher mean scores in Math (70.0), Reading (71.6), and Writing (70.8), whereas students on free or reduced lunch average lower outcomes (Math: 58.9, Reading: 64.6, Writing: 63.0).



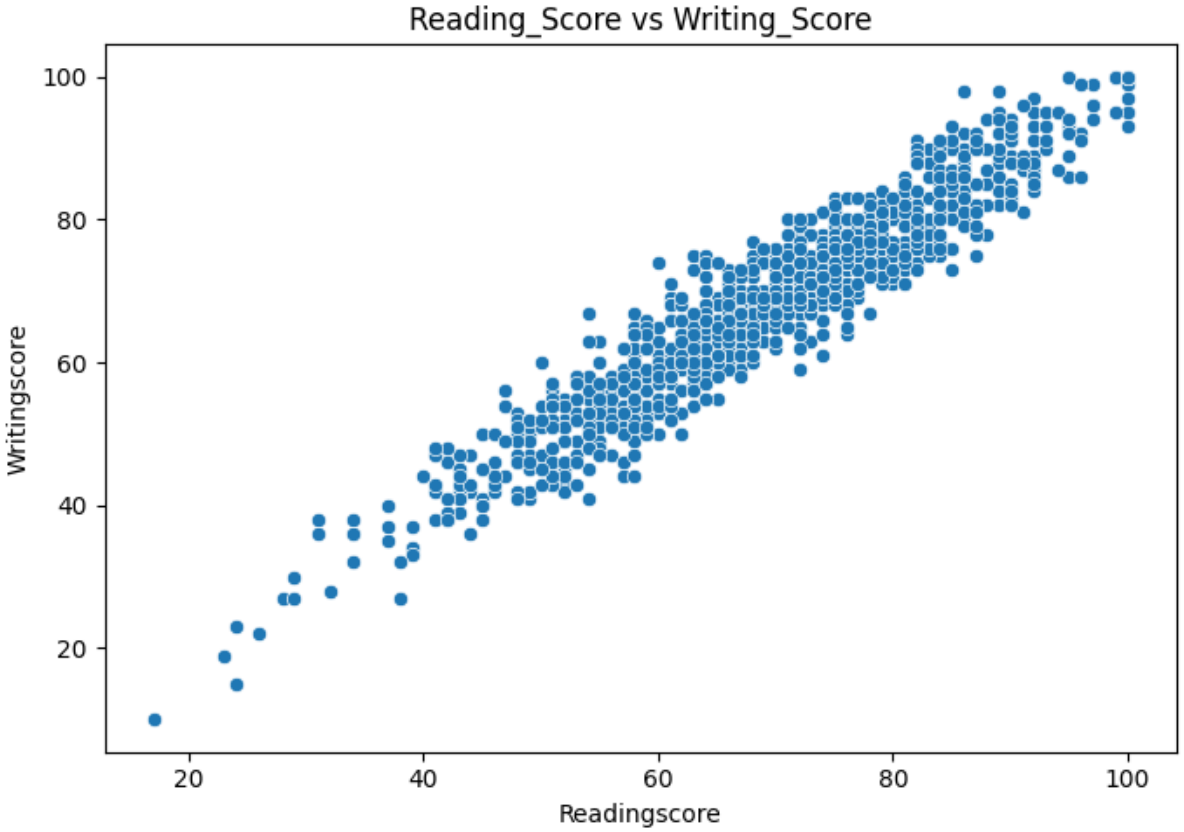
3.Test Preparation Analysis Participation in targeted test preparation courses yields a clear upward shift in performance distributions. A comparative analysis of median values and score variance shows that course completion reduces low-end performance outliers and raises median scores in Reading and Writing by nearly 8–10 points, with significant improvements in Math as well.

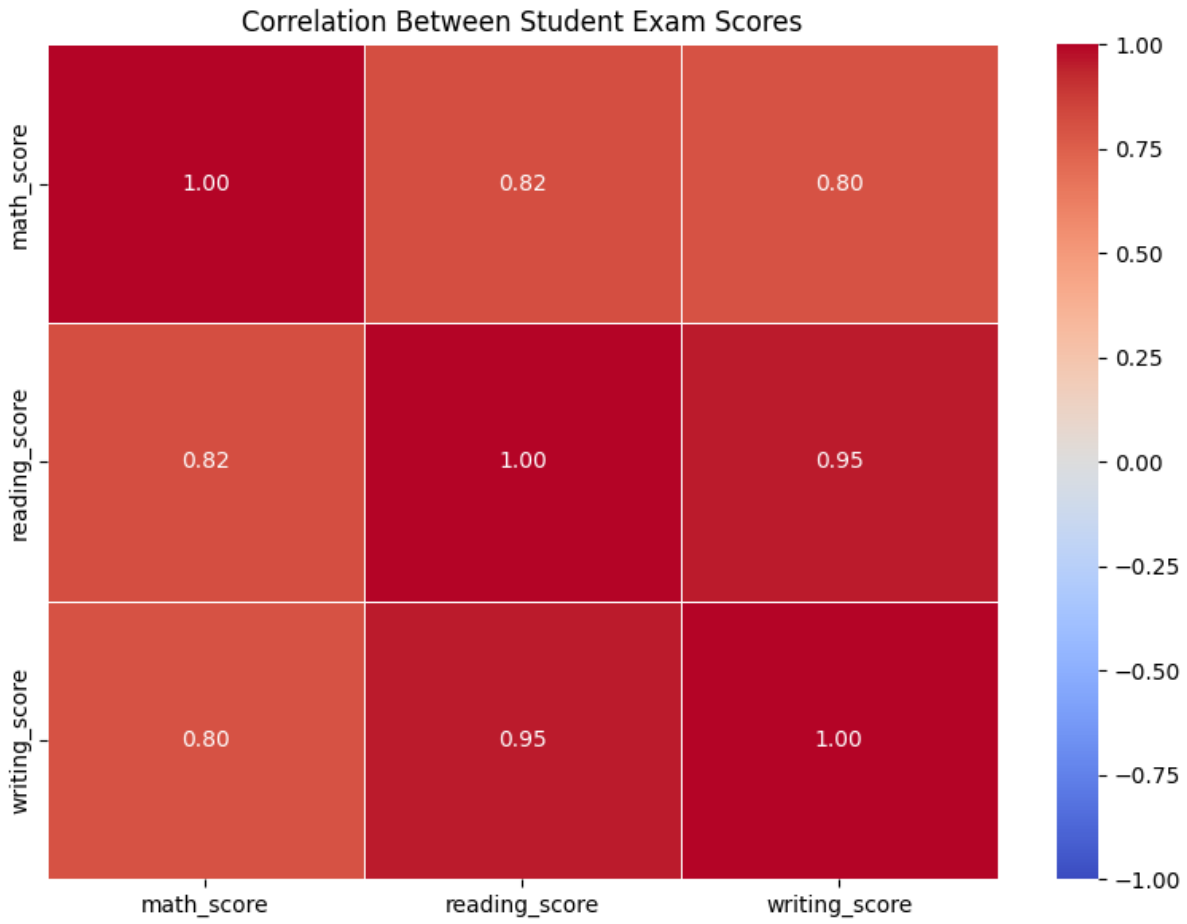






4.Subject Correlation Analysis Evaluation of subject-to-subject correlation coefficients reveals an exceptionally strong linear relationship between Reading and Writing scores ($r = 0.95$), suggesting shared underlying verbal comprehension and literacy skill sets. Correlation between Math and verbal subjects remains strong ($r \approx 0.80$), confirming overall academic alignment across disciplines.


Reading_Score vs Writing_Score
Writingscore
Readingscore
100
80
60
40
20
20
40
60
80
100



6. EDUCATIONAL EQUITY RECOMMENDATIONS

Based on the student performance analysis, some recommendations can be suggested to improve students' academic results.

1. **Additional Academic Support:**
   Students who score low marks should be provided with extra classes and individual guidance.
2. **Test Preparation Resources:**
   All students should get equal access to study materials, practice tests, and learning resources.
3. **Subject-Wise Support:**
   Special attention should be given to subjects in which students show lower performance.
4. **Equal Opportunities:**
   Educational support should be provided equally to all students without any discrimination.
5. **Performance Monitoring:**

Students' marks should be regularly monitored to identify differences in academic performance.

6. **Evaluate Improvements:**
   The effectiveness of educational support programs should be checked regularly to understand whether students' performance has improved.

7. LIMITATIONS

Every data analysis has some limitations. The following are the limitations of this Students Performance Dataset:

1. **Limited Information:**
   The dataset contains only basic student details and exam scores.
2. **Limited Subjects:**
   The analysis focuses only on Mathematics, Reading, and Writing subjects.
3. **No Personal Factors:**
   Personal interests, study habits, attendance, and learning abilities are not included.
4. **Limited Dataset:**
   The results are based only on the available dataset and may not represent all students.
5. **No Future Prediction:**
   The analysis shows existing student performance but cannot guarantee future academic results.
6. **Other Factors:**
   Family background, teaching methods, and school facilities may also affect student performance, but they are not fully covered.

Conclusion

These limitations should be considered while interpreting the results. More detailed data can help provide better and more accurate insights into student performance.